

Lecture 2:

Recap: * Periodic functions $\Rightarrow$ trigonometric interpolation with equispaced nodes

* Interpolation errors $n^{-(k-1)}$ for C^k functions
 e^{-an} for analytic functions

* Non-periodic functions $\Rightarrow$ Use polynomial interpolation instead

* However, equispaced nodes bad!
 $\exists$ analytic functions for which polynomial interpolant at equispaced points does not converge

* Nodes are the problem since
 $\inf_{p \in P_n} \sup_{[-1,1]} |f-p| \sim n^{-(k-1)}$ for C^k functions
 e^{-an} for analytic functions

Can we leverage trigonometric interpolation in some way?

Consider the change of variables $x = \cos(t)$ & let

$$g(t) = f(\cos(t))$$

Then g is 2π periodic & as smooth as f . So let's instead

use the equispaced trig interpolant of g instead

$$\tilde{g}(t) = \sum_{m=-n}^n c_m e^{imt} \quad \text{with} \quad c_m = \frac{1}{2n+1} \sum_{j=-n}^n g(t_j) \exp\left(-\frac{2\pi i j m}{2n+1}\right)$$

Symmetry of g ; i.e. $g(t) = g(-t) \Rightarrow c_m = c_{-m}$ & so

$$\tilde{g}(t) = c_0 + \sum_{m=1}^n 2c_m \cos(mt) \quad \& \quad c_m = \frac{1}{2n+1} \left(g(t_0) + 2 \sum_{j=1}^n g(t_j) \cos\left(\frac{2\pi j m}{2n+1}\right) \right)$$

What does this mean for f ?

Interpolation points

$$x_j = \cos(t_j) = \cos\left(\frac{2\pi j}{2n+1}\right) \quad j=0, 1, \dots, n$$

& basis functions

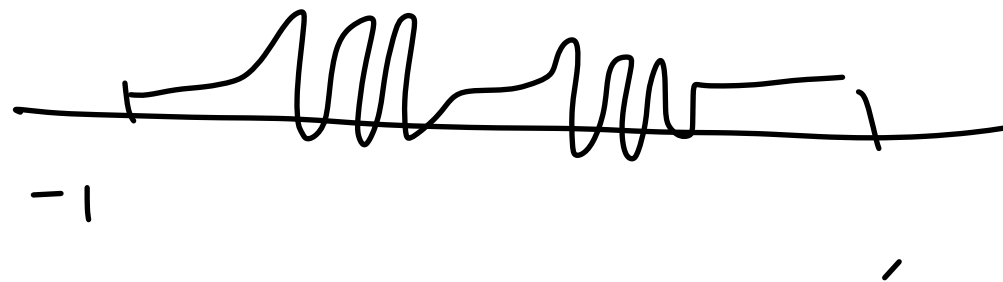
$$\cos(mt) = \cos(m \arccos(x)) = T_m(x)$$

Chebyshev polynomials

& all convergence results from Fourier land just follow

What if the function does interesting things only on part of the domain?

Chebyshev nodes cluster at end points but are more or less equispaced in the middle parts of the interval. Using high order interpolation is wasteful



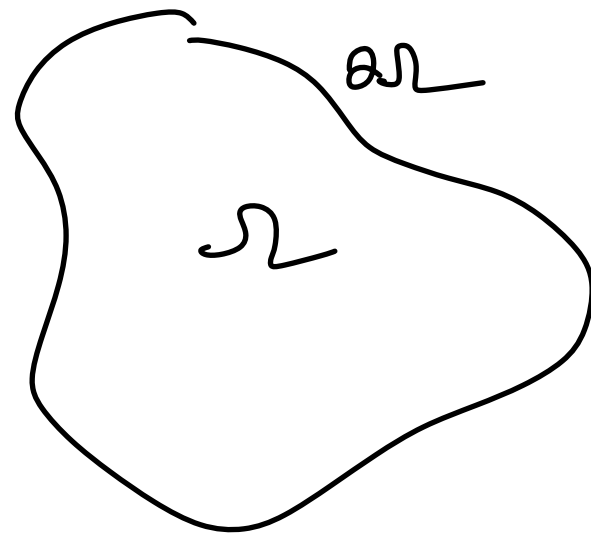
Idea: Use fixed order interpolants but on an adaptive grid of intervals.

Algorithm : 1. Construct say $p=16$ order Chebyshev expansion on $[-1, 1]$
2. Estimate the error
3. If $\text{error} < \text{tol} \rightarrow \text{Done} :)$
4. If not split into 2 intervals $[-1, 0], [0, 1]$
 & construct 2 different Chebyshev expansions
 & repeat from step 2 until the error is small on all intervals.

Boundary integral equations for Laplace's equation

Interior Dirichlet problem (IDP)

$$\begin{aligned}\Delta u &= 0 && \text{in } \Omega \\ u &= f && \text{on } \partial\Omega\end{aligned}$$



Recall that Green's function G such

$$\Delta G(x, y) = -\delta_{x=y}$$

$$G(x, y) = -\frac{1}{2\pi} \log |x - y|$$

Representation/Ansatz for u (using charges)

$$u := S[\sigma](x) = \int_{\partial\Omega} G(x, y) \sigma(y) ds_y$$

For any σ (sufficiently nice) $\Delta u = 0$ in Ω (or $\mathbb{R}^2/\Omega$)

So need σ so that boundary conditions are satisfied, i.e.

$$\lim_{x \rightarrow x_0 \in \partial\Omega} S[\sigma](x) = S[\sigma](x_0) = \boxed{\int_{\partial\Omega} G(x_0, y) \sigma(y) ds_y = f(x_0)} \quad \forall x_0 \in \partial\Omega \quad (2)$$

So if σ solves (2) then $u = S[\sigma]$ solves IDP

Boundary integral equation for σ

Integral equations

$$K[\sigma](x) = \int K(x,y) \sigma(y) dy = f(x) \quad \text{First Kind integral eq'}$$

The operator K is compact

Easy to represent on a computer because they can be well approximated by matrices (in an appropriate basis)

$\exists K_n$ (finite rank or has a matrix rep) such that $K_n \rightarrow K$

For integral operators of mathematical physics, K singular as $x \rightarrow y$ but smooth otherwise

K compact if $|K(x,y)| \leq C|x-y|^\alpha$ for some $\alpha > -1$

Other properties of compact operators

Have eigenvalues $\{\lambda_n\}_{n=0}^\infty$ and $\lambda_n \rightarrow 0$ as $n \rightarrow \infty$

(Spectrum can be a lot more complicated in infinite dimensions)

Bad news though evs of K' will approach 0

Integral operators

Better operators $(I+K)$ with K compact, its eigenvalues will approach 1 & if -1 is not an eigenvalue of K then $(I+K)^{-1}$ exists & is bounded
(Second kind integral operators)

Many results from Matrix land apply:

e.g. $\{ Ax = 0 \Rightarrow x = 0 \} \Rightarrow \exists! x \forall b$
satisfying $Ax = b$

Uniqueness $\Rightarrow$ existence

$N(I+K)$ is finite dim & $N(I+K) = N(I+K^*)$

moreover $R(A) = N(A^*)^\perp$ & $N(A) = R(A^*)^\perp$

These things seem obvious in finite dim but operators on function spaces can be a lot more complicated

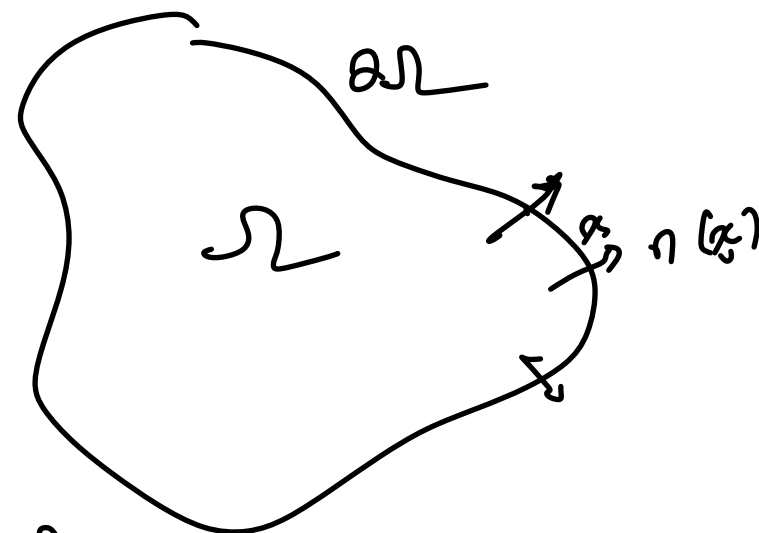
Back to Laplace's equation.

$$\begin{array}{ll} \Delta u = 0 & \text{in } \Omega \\ u = f & \text{on } \partial\Omega \end{array}$$

Representation

$$u = \int_{\partial\Omega} G(x, y) \sigma(y) dy$$

gives first kind IE for σ



But other representations available, can use any number of derivatives of Green's functions.

Let's try dipoles

$$u(x) = \int_{\partial\Omega} n(y) \cdot \nabla G(x, y) \sigma(y) ds_y \quad D[\sigma](x)$$

$$= \frac{1}{2\pi} \int_{\partial\Omega} \frac{(x-y) \cdot n(y) \sigma(y) ds_y}{|x-y|^2}$$

$$= \operatorname{Re} \left(\frac{1}{2\pi i} \int_{\partial\Omega} \frac{\sigma(\xi)}{z-\xi} d\xi \right)$$

$$z = x_1 + i x_2$$

$$\xi = y_1 + i y_2$$

Laplace equation IDP

$$u(x) = P[\sigma](x) = \operatorname{Re} \left(\frac{1}{2\pi i} \int_{\partial\Omega} \frac{\sigma(\xi)}{z-\xi} d\xi \right)$$

Clearly u still satisfies the PDE for any σ because derivatives of Green's functions satisfy the PDE away from the source

$$\text{now } \lim_{x \rightarrow x_0} u(x) = -\frac{1}{2} \sigma(x_0) + \text{p.v.} \frac{1}{2\pi} \int \frac{(x_0-y) \cdot n(y)}{|x-y|^2} \sigma(y)$$

$$= \left(-\frac{I}{2} + \underbrace{D^{\text{p.v.}}}_{\text{compact operator}} \right) [\sigma](x_0)$$

$$\forall x_0 \in \partial\Omega$$

So if σ solves

$$\left(-\frac{I}{2} + D^{\text{p.v.}} \right) \sigma = f \quad (*)$$

$$\& u = D[\sigma]$$

then u solves IDP

SKIE so uniqueness $\Rightarrow$ existence

compact operator

HW: Show that

$$K(x,y) = \frac{(x-y) \cdot n(y)}{|x-y|^2}$$

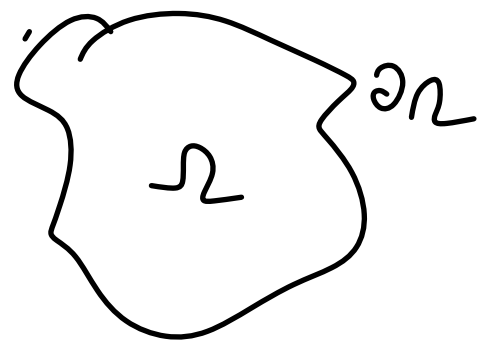
Bounded $\forall x,y \in \partial\Omega$

Thm: $\forall f \exists ! \sigma$ satisfying $(*)$

Interior Neumann problem

$$\Delta u = 0 \quad \text{in } \Omega$$

$$\frac{\partial u}{\partial n} = f \quad \text{on } \partial\Omega$$



Q: Is there always a solution?
 A: No, solution only exists as long as

$$\int_{\partial\Omega} f = \int_{\partial\Omega} \frac{\partial u}{\partial n} = \int_{\Omega} \Delta u = 0$$

Moreover, solution only determined upto a constant

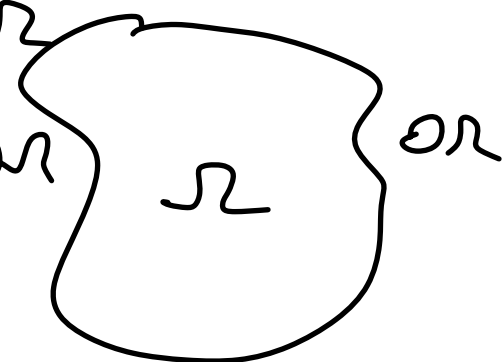
Fix: choose some normalization

Integral representation:

$$u = S[\sigma](x)$$

$\Delta u = 0$ in Ω

$\frac{\partial u}{\partial n} = f$ on $\partial\Omega$



$$\lim_{h \rightarrow 0^+} \nabla u(x_0 - h n(x_0)) \cdot n(x_0)$$

$$= \frac{1}{2} \sigma(x_0) - p.v. \frac{1}{2\pi} \int \frac{(x_0 - y) \cdot n(x_0) \sigma(y) dy}{|x_0 - y|^2} \leftarrow S'[\sigma](x_0)$$

$$= f(x_0)$$

$$\left(\frac{I}{2} + S' \right) \sigma = f$$

on $\partial\Omega$

HW: Show that

$$\int_{\partial\Omega} u(x) S'[\sigma](x) dx = \int_{\partial\Omega} \sigma(x) D^p v[u](x) dx$$

on

i.e. S' and D are adjoints of one another

Note different sign of identity

Integral eqⁿ is second kind, so are we done?

Unfortunately no! If σ is a constant on $\partial\Omega$

$$\text{then } \frac{1}{2} \sigma + D^p v \sigma = 0 \text{ on } \partial\Omega$$

Recall $N(A) \subseteq N(A^*)$

so $\left(\frac{I}{2} + S' \right)$ also has a null vector

$$\left(\frac{\mathbb{I}}{2} + S'\right)\sigma = f$$

$$\forall \sigma \quad \int \mathbb{1} \left(\frac{\mathbb{I}}{2} + S'\right)\sigma = \int \sigma \left(\frac{\mathbb{I}}{2} + D^{PV}\right)[\mathbb{1}] = 0$$

But $\int f \neq 0$ required for existence of solⁿs to PDE

So as long as data is consistent we are fine

How do we fix the null space? Add something that's not in the range!

$$\left(\frac{\mathbb{I}}{2} + S'\right)\sigma + \frac{1}{|\partial\Omega|} \int_{\partial\Omega} \sigma \, ds_n = f \quad \text{is an invertible SKIE for } \sigma$$

$$\int \sigma = \int \left(\frac{\mathbb{I}}{2} + S'\right)\sigma + \int \frac{1}{|\partial\Omega|} \int_{\partial\Omega} \sigma = \cancel{\int f} = 0$$

$$\Rightarrow \int_{\partial\Omega} \sigma = 0$$

So if σ solves $\left(\frac{\mathbb{I}}{2} + S'\right)\sigma = f$ it also solves $\int \sigma = 0$